

Exercise Sheet 6

Explainability

Hint: From now on, you may be asked to present your solution in digital form to the class. Please bring a laptop and an HDMI adapter so you can connect to the projector. If you are unable to do so, submit your solutions to kirchhei@ovgu.de prior to the exercise. Solutions submitted after the exercise has started will not be accepted.

Explainability Concepts

Exercise 6.1: Why Explainability?

optional

What are advantages of explainable models in a safety-critical context? What are potential disadvantages or limitations of current explainability methods?

1. Advantages in a Safety-Critical Context

- **Verification of Correct Logic:** It ensures the model is making decisions based on actual causal features (e.g., detecting a pedestrian by their shape) rather than spurious correlations (e.g., detecting a pedestrian because of the color of the sky).
- **Debugging and Auditing:** When an autonomous system fails, explainability helps safety engineers trace the root cause of the failure, allowing them to patch or retrain the system effectively.
- **Regulatory Compliance:** Many high-risk deployment domains (such as autonomous driving, healthcare, or aerospace) legally mandate a "right to explanation" to ensure accountability.

2. Disadvantages and Limitations

- **The Fidelity-Interpretability Trade-off:** Post-hoc explanation methods are often approximations of a highly complex model. They might present a clean, understandable summary that doesn't actually reflect the true, messy internal mechanics of the network.
- **Vulnerability to Manipulation (Fairwashing):** Explanations can be easily fooled or optimized adversarially. A model can be trained to produce "good-looking" explanations to mask biased, unsafe, or faulty internal reasoning.
- **Automation Bias:** If an explanation looks highly convincing and intuitive, human supervisors are more likely to blindly trust the AI's final output, even when that output is completely wrong.

Exercise 6.2: Local vs. Global Explainability

Describe the difference between *local* and *global* explainability. Name one method for each kind and explain what question it answers.

The Difference

- **Local Explainability** focuses on explaining a **single, individual prediction**. It zooms in on one specific input to show why the model made that exact choice.
- **Global Explainability** focuses on explaining the **overall behavior of the entire model** across a whole dataset. It zooms out to reveal the overarching rules, circuits, or features the model prioritizes across all inputs.

Type	Example Method	Question It Answers
Local	SHAP (SHapley Additive exPlanations) or LIME	<i>"Why did the model classify this specific image as containing a pedestrian?"</i>
Global	Permutation Feature Importance	<i>"Across the entire dataset, which features are generally the most important to the model's performance?"</i>

Exercise 6.3: Saliency vs. Occlusion

Briefly describe the saliency method and the occlusion method. What is one advantage and one disadvantage of each compared to the other?

- **Saliency Method (Gradient-Based):** Calculates the gradient of the model's output score with respect to each input pixel. It highlights which pixels, if slightly altered, would change the output prediction the fastest.
- **Occlusion Method (Perturbation-Based):** Systematically slides a gray/blank masking box across the input image and measures how much the model's prediction confidence drops when certain regions are blocked.

Direct Comparison

- **Saliency Maps**
 - *Advantage: Extremely fast.* It only requires a single backward pass through the neural network to generate the map.
 - *Disadvantage: Noisy and saturated.* Gradients can "shatter" or saturate, meaning a crucial feature might have a gradient of zero simply because the model is already 100% confident, making the resulting map look like random noise.
- **Occlusion Maps**
 - *Advantage: True causal measurement.* It directly tests how the model behaves when information is completely removed, making it highly intuitive and reflective of feature necessity.
 - *Disadvantage: Computationally expensive.* You have to pass the image through the model hundreds of times as the patch slides across the screen, which scales poorly with high-resolution data.

Exercise 6.4: Chain-of-Thought Fidelity

Modern LLMs often produce a thinking trace (chain of thought) alongside their answer.

1. What does it mean for a thinking trace to be *faithful*? Why is faithfulness hard to verify?

Faithfulness and Why It's Hard to Verify

- **Definition:** A thinking trace is *faithful* if the generated step-by-step logic accurately represents the actual internal reasoning process the LLM used to compute its answer.
- **Why it's hard to verify:** LLMs generate tokens sequentially. The model can easily generate a beautiful, human-sounding rationalization (a post-hoc justification) for an answer it actually arrived at via entirely different, hidden shortcut heuristics in its weights.

2. What is *simulatability* as a measure of thinking-trace quality? Give an example scenario where simulatability is satisfied but the trace is still unfaithful.

Simulatability and the Trace vs. Output Mismatch

- **Simulatability:** A measure of quality indicating whether a human reader can look exclusively at the thinking trace and accurately predict what final answer the model will produce.
- **Unfaithful Scenario:** Imagine a math-solving LLM that has a hidden sycophantic bias to agree with the user. If the user asks "*Is $2+2=5$?*", the model might output a flawless, mathematically sound thinking trace explaining that $2+2=4$ (satisfying *simulatability*, because anyone reading it expects the answer to be 4). However, at the very last token, it outputs: "*Therefore, the answer is 5.*" The trace was highly simulatable, yet entirely unfaithful to the final output token choice.

3. What does *counterfactual simulatability* add? Why is it a stricter test?

Counterfactual Simulatability

- **What it adds:** It measures whether **modifying** a step in the thinking trace causes a predictable, corresponding shift in the final output.
 - **Why it's stricter:** It tests for a *causal link* rather than just a correlation. If you manually change a line in the model's calculation from $x=5$ to $x=10$, a truly faithful model's final answer must adapt to match that intervention. If the output remains completely unchanged, the trace is a fabrication.
4. Why does an unfaithful thinking trace pose a safety risk? Give one concrete example.

Safety Risks of Unfaithful Traces

- **The Risk: Deception and Hidden Failure Modes.** If an operator reviews a trace and sees flawless reasoning, they will trust the output. This creates a false sense of security, masking malicious intent, hallucinations, or dangerous system errors.
- **Concrete Example:** A medical LLM is asked to calculate a patient's medication dosage. It outputs a perfectly logical, safe trace detailing the calculation step-by-step. However, due to an unfaithful trace error or an adversarial prompt attack, the final text generated specifies a lethal dose. The supervising doctor reads the logical trace, assumes it's correct, and signs off on a dangerous prescription.

Practical: Explaining the CARLA Model

Exercise 6.5: Applying an Explainability Method

Choose one local explainability method from the lecture: occlusion, CAM, Grad-CAM, or saliency maps. Apply it to each of your three CARLA models.

1. Briefly describe how your chosen method works and why you chose it.
2. Select five correctly classified images - at least one per model. Visualize the explanation (heatmap or sensitivity map) overlaid on the original image.
3. Do the highlighted regions correspond to the relevant objects (e.g., does the map highlight the pedestrian when the model predicts “pedestrian present”)?
4. Select three misclassified images and apply the same method. Do the explanations give any insight into why the model failed?

Exercise 6.6: Explainability as a Diagnostic Tool

1. Suppose your explanation method reveals that your model predicts “pedestrian present” based primarily on a region of the sky rather than the pedestrian itself. What would this imply about the model’s generalization? What could cause this behavior?
2. Now test your models on images from different conditions than the ones they were trained on - for example a different town layout, nighttime, or foggy weather. Apply the same explanation method as before.
 - (a) Do the highlighted regions still correspond to the relevant objects?
 - (b) Do you observe evidence that the model relied on background or spurious features (lighting, road texture, sky color) that are specific to the original training conditions?
 - (c) How do the model’s accuracy and explanation quality change across conditions?

